VALUATION SCHEME

QPCODE: 25900905

MAHATMAGANDHIUNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/ IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCPEG100 – FUNDAMENTALS OF PETROLEUM GEOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A

Multiple Choice Questions Answer All Questions.

Each question carries one mark. (10 × 1= 10)

1. What role do cap rocks play in petroleum reservoirs? [K] / [CO1]
   b, they trap hydrocarbons and prevent their escape
2. What is thermal alteration in the context of petroleum? [U] / [CO1]
   b, The chemical transformation of hydrocarbons under heat
3. What is the main hydrocarbon produced during the peak catagenesis phase? [U] / [CO2]
   b, Methane
4. Which of the following is type of structural trap? [K] / [CO2]
   a, Fault trap
5. Identify the method uses satellite imagery to analyze the earth's surface for signs of oil and gas potential. [K] / [CO3]
   b, Photogeology
6. Label the type of trap commonly identified using gravity and seismic methods in petroleum exploration. [U] / [CO3]
   c, Structural trap
7. In borehole logging, which of the following techniques is used to detect fractures in rock formations? [U] / [CO3]
   c, Acoustic logging
8. Name the gas primarily found in gas hydrates. [K] / [CO4]
   b, Methane
9. Sour crude oil is primarily characterized by: [K] / [CO4]
   c, High sulphur content
10. Refractive index important in petroleum products: [U] / [CO5]
    b, To assess purity and composition

Part B

Short Answer Type Questions Answer any **4** questions.

Each question carries **3** marks. (4 × 3= 12)

(Give Full credit to any 3 Points corresponds to each questions)

11. Explain the process of catagenesis and its important in hydrocarbon generation. [U] / [CO1]

- Catagenesis is the intermediate stage in the transformation of buried organic matter, where increasing heat and pressure (about 60–120°C) break down kerogen through thermal cracking. This temperature range is called the oil window, where kerogen decomposes into liquid hydrocarbons (oil) and gaseous hydrocarbons (natural gas). **(1.5 marks)**
  - It is highly significant because different types of kerogen generate varying amounts of oil and gas, and the efficiency of catagenesis determines how much hydrocarbon a source rock can produce. The process also influences the physical and chemical properties of the hydrocarbons, affecting their migration and accumulation in reservoir rocks. Thus, catagenesis is a key stage directly linked to economic hydrocarbon generation. **(1.5 marks)**

12. Describe the transition of oil window to gas window. [U] / [CO2]
    The transition from oil window to gas window occurs as the subsurface temperatures rise beyond 120°C. In the oil window (60–120°C), kerogen mainly breaks down to form liquid hydrocarbons. When temperatures exceed 120°C, the process enters the gas window (about 120–200°C), where remaining kerogen and previously formed oil undergo further thermal cracking to produce natural gas, mainly methane. At even higher temperatures, the capacity for hydrocarbon generation declines, and organic matter may decompose into non-hydrocarbon residues. This transition marks the shift from oil formation to dominant gas production during burial and heating. (**Any 3 Points – 3 marks**)
13. Compile electromagnetic methods with seismic surveys in oil exploration techniques. [A]/[CO3]
    Electromagnetic (EM) methods and seismic surveys are complementary geophysical techniques used in petroleum exploration. EM methods investigate subsurface electrical conductivity, helping detect high-resistivity hydrocarbon-bearing formations, map salt domes, and identify fluid distributions—especially useful in areas where seismic data are less reliable. In contrast, seismic surveys use artificial seismic waves to create detailed images of subsurface structures based on differences in rock and fluid seismic properties. By analyzing reflected and refracted waves, geophysicists can locate traps, faults, folds, and other structural features. Together, EM and seismic methods improve accuracy in identifying potential hydrocarbon reservoirs. (**Any 3 Clear Points – 3 marks**)
14. Discuss the importance of remote sensing and Geographic Information System (GIS) in petroleum exploration. [U] / [CO3]
    Remote sensing is important in petroleum exploration because it uses satellite images, aerial photos, and drone data to identify surface features such as folds, faults, lineaments, and areas affected by hydrocarbon seepages. It helps in mapping regional geology, detecting surface anomalies, and selecting potential drilling sites in a cost-effective way.
    Geographic Information System (GIS) complements this by integrating and analyzing various datasets—remote sensing data, seismic results, well logs, and geochemical information—on a single platform. GIS helps map hydrocarbon prospects, identify structural traps, and evaluate environmental and logistical factors. Together, remote sensing and GIS improve exploration accuracy, reduce risks, and support efficient planning of petroleum exploration activities. (**Any 2 Points from each part Give full credit**)

15. Explain the significance of understanding the composition of oxygen compounds in crude oil. [U] / [CO4]

    Understanding the composition of oxygen compounds in crude oil is important because these components—such as alcohols, phenols, and ketones—affect key properties like viscosity, volatility, and overall crude quality. Excessive oxygen content can make crude oil unstable by promoting oxidation, leading to the formation of acids, gums, and deposits that corrode equipment and clog pipelines. Knowing the types and amounts of oxygen compounds helps refineries manage storage conditions, use appropriate additives, and prevent degradation, ensuring efficient processing and safe handling of crude oil. (**Any 3 Points – Give full credit**)

16. Explain how does pour point impact the usability of fuels? [U] / [CO5]

    The pour point is the lowest temperature at which a fuel remains fluid and can flow. It directly impacts fuel usability, especially in cold climates. Fuels with a high pour point may thicken or solidify at low temperatures, causing flow problems in pipelines, storage tanks, and engines. This can lead to reduced efficiency, ignition issues, and equipment damage. Therefore, understanding and controlling the pour point is essential to ensure smooth handling, transport, and performance of fuels under varying temperature conditions. (**Pour point definition 1 mark, explanation 2 marks**)

Part C

Short Essay Type Questions

Answer any **4** questions. Each question carries **7** marks. (7 × 4= 28)

17. Justify that modern theory provides better explanation for origin of petroleum than other theories. [E] / [CO1]

    The modern theory for the origin of petroleum, known as the biogenic theory, posits that petroleum is primarily derived from the decay of ancient marine organism such as plankton and algae, which were buried under layers of sediment millions of years ago, undergo anaerobic decomposition by bacteria in which the fat content from the sediments gets converted into fatty acids. These fatty acids undergo further transformation to form kerogen. Due to combined action of heat and pressure, the saprophytic kerogen converted into oil and gas. This oil and gas undergo 1° migration from fine grained rocks to coarse grained sedimentaryrocks. 2° migration of this fluid leads to accumulation which result in formation of oil wells.

Key reasons that make it a more compelling explanation for the origin of petroleum compared to alternative theories

- The biogenic theory is supported by extensive chemical and geological data.
- Fossils of ancient marine microorganisms are found in petroleum-rich formations.
- Laboratory experiments also supports the theory, Sedimentary structures like hydrocarbon traps and reservoirs are as easily explained by the modern -biogenic theory

(**Description of Modern Theory 2 Marks, Justification 5 Marks)**

18. Explain the differences between proved, probable, and possible reserves in petroleum exploration. [U] / [CO2]

(Any 2 points each for proved, probable, and possible reserves Give Full credit)

Proved, probable, and possible reserves represent different levels of confidence in the recoverability of petroleum resources and are essential for evaluating the economic potential of an oil or gas field. Proved reserves are the most reliable estimates. They refer to quantities of petroleum that can be recovered with reasonable certainty from known reservoirs under current economic conditions, operating methods, and government regulations. In deterministic methods, this means a high degree of confidence, while in probabilistic terms, there is at least a 90% probability of recovery. Proved reserves are further subdivided into developed (ready for production) and undeveloped (requiring additional drilling or infrastructure).

Probable reserves fall under unproved reserves but have a higher likelihood of being recovered compared to possible reserves. They rely on geological and engineering data but lack the certainty of proved reserves due to technical or economic uncertainties. Probable reserves are considered more likely than not to be recoverable, typically with at least a 50% probability. However, they may not be extracted if economic conditions make recovery unfeasible.

Possible reserves represent the least certain category among the three. They are also unproved reserves but have a lower chance of successful extraction compared to probable reserves. Geological and engineering data indicate that they are less likely to be recovered, with a probability of at least 10%. These reserves often depend on future technological improvements or changes in economic conditions to become viable.

The key difference lies in the degree of confidence: proved reserves offer the highest certainty of commercial recovery, probable reserves indicate moderate confidence, and possible reserves reflect low confidence, making them the most uncertain category in petroleum exploration.

19. Compare the electrical and resistivity methods in subsurface exploration. How do these techniques work, and in what contexts are they most effective? [An] / [CO3]

(Explanation on electrical and resistivity method 5 marks and answer to most effectiveness of the method 2 marks)

The electrical method measures the general electrical resistivity of subsurface rocks. Hydrocarbons are poor conductors, so formations containing oil or gas show higher resistivity compared to water-saturated rocks. This method uses electrodes placed on the surface or in boreholes to pass current and measure potential differences. It helps map subsurface structures and locate possible reservoirs. The method is simple, low-cost, and useful in areas where seismic surveys are difficult, though it has a limited depth of investigation and can be influenced by soil moisture, clay content, and surface conductivity.

The resistivity method**,** a more specific technique within electrical surveys, focuses on measuring the electrical resistance of rocks to identify hydrocarbon zones. Since hydrocarbons increase resistivity, this method is widely used in well logging and surface surveys to distinguish between hydrocarbon-bearing and water-bearing formations. It also helps estimate formation thickness and porosity. Although it is non-invasive and effective, its accuracy can be affected by the salinity of formation water and clay content, and it requires careful calibration, especially in heterogeneous formations.

Overall, electrical methods are most effective for broad subsurface mapping and identifying resistivity contrasts near the surface, while resistivity methods are particularly valuable in well logging and detailed reservoir characterization. Together, they provide essential information for identifying and evaluating potential hydrocarbon reservoirs.

20. Relate geobotanical techniques with other surface exploration methods like geochemical or geological exploration. [U] / [CO3]

(Give 2.5 marks each for three points of geobotanical and geochemical method and for geological exploration give 2 marks)

Geobotanical exploration is an important surface exploration technique that complements other methods such as geochemical and geological exploration in petroleum prospecting. Geobotanical methods rely on studying vegetation type, distribution, and health to detect anomalies caused by natural hydrocarbon seepages. Certain plants absorb trace hydrocarbons or show stress when exposed to altered soil chemistry, making them useful indicators of subsurface petroleum. Like geochemical exploration, geobotanical techniques are non-invasive, cost-effective, and useful for early-stage identification of prospective zones. Both methods provide indirect evidence of hydrocarbons and help narrow down targets before expensive drilling or geophysical surveys. Geochemical methods analyze soils, rocks, gases, and water for hydrocarbon microseepages, while geobotanical methods focus on plant responses. Both techniques may be affected by environmental factors, leading to false positives or negatives, highlighting the need for integration with other surveys. Compared to geological exploration**,** which examines rock types, structural features, folds, faults, and sedimentary basins, geobotanical surveys offer an advantage in vegetation-covered or remote regions where rock exposures are limited. Geological methods provide essential information about basin structure but are restricted to surface features and may not reveal deeper reservoirs. Geobotanical indicators help refine geological interpretations by pointing to potential seepage zones.

21. Explain the impact of olefins on the refining process of crude oil. [U] / [CO4]

(Any 5 points give 7 marks)

Olefins are unsaturated hydrocarbons, which means the double bond is present between the two carbon atoms in the formula. The generic formula is (C_nH_{2n}), and the lowest member of this homologous series is ethylene, C_2H_4.This series is known as alkenes. These are highly reactive and can react to themselves to mono olefins. Olefins are not present in crude oil, but they are produced by thermal and catalytic decomposition or dehydrogenation of normal paraffins. Olefins can complicate the refining process because their double bonds make them more reactive, which can lead to polymerization or other unwanted reactions. They are often targeted for conversion into more stable compounds during refining. Olefins are generally undesirable in finished products because, the double bonds are reactive and the compounds are more easily oxidized and polymerized to form gums and varnishes, so they can be removed by absorption in sulfuric acid.The reactivity of olefins makes them valuable feedstocks in the petrochemical industry, particularly in the production of polymers, synthetic fibers, and chemicals.

22. Discuss the significance of density in characterizing hydrocarbons and its implications for various industries. [U] / [CO5]

(Give 3 marks for explanation on density characterization and 4 marks to application)

Density is a fundamental physical property used to characterize hydrocarbons because it directly reflects their composition, molecular structure, and behaviour under different conditions. Defined as mass per unit volume, density changes significantly with temperature. As temperature increases, molecular kinetic energy rises, causing hydrocarbons to expand and their density to decrease. This temperature–density relationship is crucial for accurate measurement and handling of petroleum products. In the oil and gas industry, understanding density is essential during production, transportation, and refining. Pipelines must be designed to accommodate density variations caused by temperature fluctuations to maintain smooth flow and prevent issues such as phase separation or blockages. Density also influences reservoir evaluation, fluid modeling, and metering systems used to calculate volumes and economic value.

In refining**,** density plays an important role in separation processes such as distillation, where hydrocarbon fractions are separated based on boiling points related to molecular weight and density. Accurate density measurement ensures product quality and helps classify fuels like gasoline, diesel, and kerosene. Hydrometer method is commonly used for determining density.
Density has implications for transportation and storage logistics**,** influencing tank design, material selection, and flow assurance. It is also relevant in environmental management**,** as density helps predict how hydrocarbons behave during spills, including their spreading, mixing, and biodegradation tendencies.
